Indian Institute of Technology, Kanpur

Department of SPASE Laboratory

Report Of Radio Astronomy Laboratory: SPA617

RF TRANSMISSION

by

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1. Aim

To investigate RF transmission-line behaviour by measuring voltage against frequency for an open-circuit cable and a cable terminated with 50 Ω; identify voltage maxima and minima caused by standing waves; and estimate reflection coefficient, phase velocity/velocity factor and characteristic impedance from the recorded data.

2. Apparatus Required

- RF signal generator
- Digital storage oscilloscope (DSO), input impedance set to 1 MΩ
- RF Tee connector
- Coaxial cable under test (length noted as approximately 0.92 m)
- 50 Ω termination
- Open-circuit termination
- RF cables, adapters and connectors

3. Theory

3.1 Transmission lines and standing waves

A transmission line carries an incident RF wave toward a load. If the load impedance differs from the line characteristic impedance, part of the wave is reflected. The incident and reflected waves superpose to form a standing-wave pattern. When frequency is swept, their relative phase changes and the measured voltage varies between maxima and minima.

3.2 Reflection coefficient

At a voltage maximum, incident and reflected waves add; at a minimum, they partially cancel. The reflection-coefficient magnitude can be estimated from a suitable maximum/minimum pair:

$$|\Gamma| = \frac{(Vmax - Vmin)}{(Vmax + Vmin)}$$

An ideal matched load has Γ = 0. An open circuit reflects nearly all incident voltage (|Γ| ≈ 1), creating pronounced standing-wave ripple.

3.3 Phase velocity and velocity factor

For consecutive maxima or consecutive minima on a cable of length ℓ, the frequency spacing Δf relates to propagation speed by:

$$vp = 2\ell\Delta f \text{ and } VF(\text{velocity factor}) = vp/c, c \approx 3.00 \times 10^8 m/s$$

The extrema must be consecutive. If the selected extrema are separated by multiple intervals, the frequency spacing must be divided by that interval count.

3.4 50 Ω termination and matching

A 50 Ω termination is intended to match a nominal 50 Ω coaxial cable. Matching absorbs most incident power and reduces reflected-wave amplitude and voltage ripple. The manual's procedure compares open-circuit and 50 Ω-terminated responses.

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4. Procedure

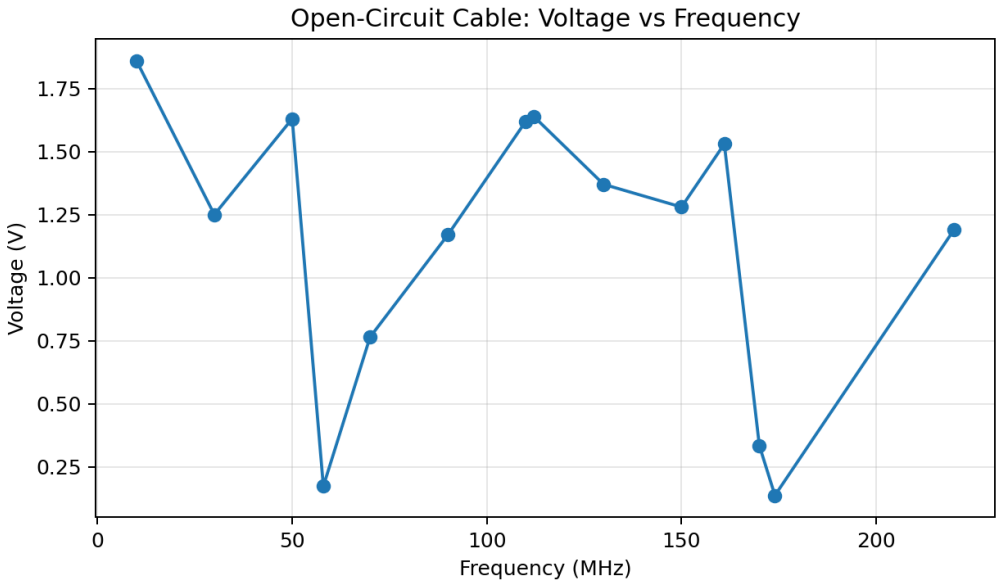
1. Measure the physical cable length; we record approximately 0.92 m.
2. Connect signal generator, Tee connector, cable under test and oscilloscope as described in the manual.
3. Set oscilloscope input impedance to 1 MΩ and keep generator amplitude constant during the sweep.
4. Leave the far end open, sweep frequency and record the voltage at the selected frequencies.
5. Plot voltage versus frequency and identify maxima and minima in the standing-wave pattern.
6. Terminate the cable with 50 Ω and repeat the voltage measurements.
7. Compare both plots and calculate the reflection/velocity quantities only using appropriate extrema and the stated formulas.

5. Observation Tables

5.1 Open-circuit cable

Frequency (MHz)	Measured voltage	Note
10	1.860 V	
30	1.250 V	
50	1.630 V	
58	173.10 mV	Minima
70	764.32 mV	
90	1.170 V	
110	1.620 V	
112	1.640 V	Maxima
130	1.370 V	
150	1.280 V	
161	1.530 V	
170	334.00 mV	
174	136.79 mV	Minima
220	1.190 V	Maxima

Figure 1. Voltage versus frequency for the open-circuit cable.



5.2 50 Ω termination

Frequency (MHz)	Measured voltage	note
62	1.441 V	
110	830 mV	Minimum
158	1.210 V	Maximum
226	637 mV	Minimum

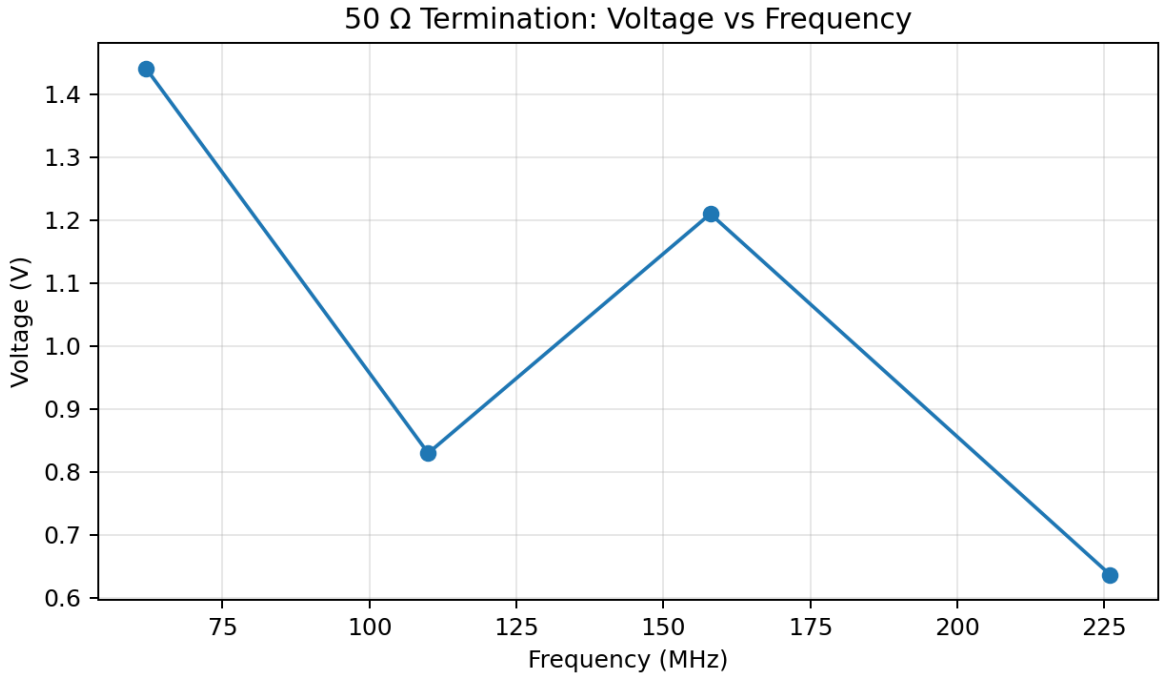
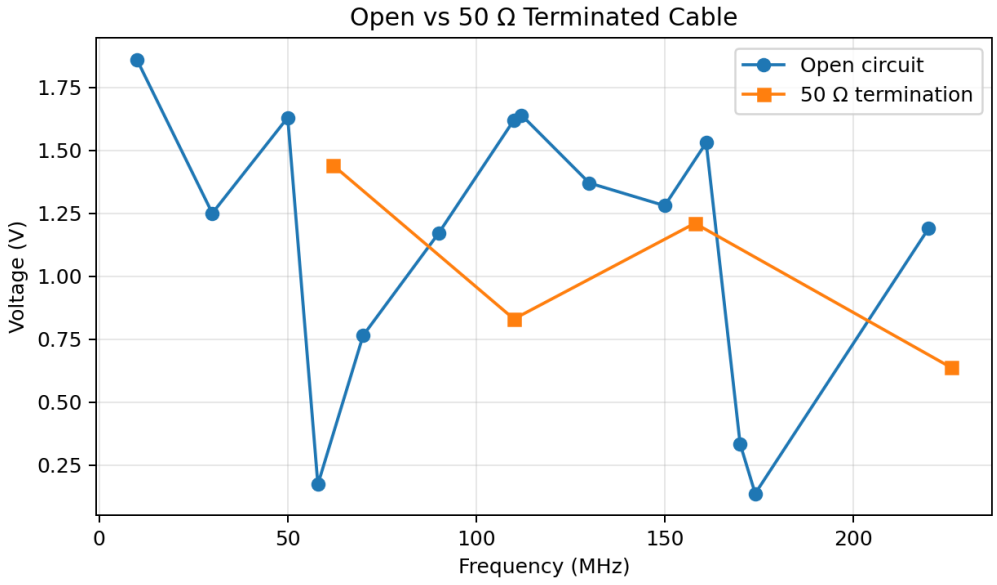


Figure 2. Voltage versus frequency with a 50 Ω termination.

Figure 3. Comparison of open and terminated measurements; sampling frequencies differ.



6. Calculations

6.1 Open-circuit reflection coefficient

Using the recorded maximum Vmax = 1.640 V and the low reading Vmin = 173.1 mV = 0.1731 V:

$$|\Gamma| = \frac{(1.640 - 0.1731)}{(1.640 + 0.1731)} = |\Gamma| = 0.809$$

6.2 Frequency spacing, phase velocity and velocity factor

We identify minima near 58 MHz and $174MHz$:

$$\Delta f = 174 - 58 = 116MHz$$

$$Usingcablelengthl = 0.92m: vp = 2 \times 0.92 \times 116 \times 10^6 = 2.1344 \times 10^8 m/s$$

$$VF = vp/c = 0.711$$

This assumes the two minima are consecutive. The data also show a low reading near 170 MHz (334 mV) and a minimum at 174 MHz (136.79 mV).

6.3 50 Ω terminated measurements

Using the recorded maximum 1.21 V and minimum 0.830 V:

$$|\Gamma| = \frac{(1.21 - 0.830)}{(1.21 + 0.830)} = |\Gamma| = 0.186$$

$Inductance\ per\ unit\ lengt \boxed{L' = \frac{Z_0}{v_p}}$

$Using\ the\ approximate\ characteristic\ impedance(Z_0 = 50\ \Omega)$

$calculated\ phase\ velocity(v_p = 2.1344 \times 10^8$ m/s So, $L' = \frac{50}{2.1344 \times 10^8} = = 2.342 \times 10^{-7}$ H/m = 234.3 nH/m

$$Capacitance\ per\ unit\ length\ \boxed{C' = \frac{1}{Z_0 v_p}}$$

$$C' = \frac{1}{50(2.1344 \times 10^8)} = 9.370 \times 10^{-11}\ \text{F/m}$$

$$\boxed{C' \approx 93.7\ \text{pF/m}}$$

7. Result

Parameter	Recorded value	Calculation / interpretation
Cable length	≈ 0.92 m	Used for phase-velocity estimate
Open-circuit maximum	1.64 V at 112 MHz	
Open-circuit low reading	173.1 mV at 58 MHz	
Open-circuit minimum	136.79 mV at 174 MHz	
Selected minima spacing	116 MHz	174 − 58 MHz
Phase velocity	≈ 2.13 × 10⁸ m/s	Calculated using vp = 2ℓΔf
Velocity factor	appears ≈ 0.77	Direct calculation gives ≈ 0.711
50 Ω case maximum	1.21 V at 158 MHz	
50 Ω case minima	830 mV at 110 MHz; 637 mV at 226 MHz	

Inductance	234.3 nH/m	
Capacitance	93.7 pF/m	
Open-circuit $\|\Gamma\|$	≈ 0.809	From 1.640 V and 0.1731 V; selected pair requires confirmation
Approximated Z_0	50Ω	

8. Discussion

8. The open-circuit sweep shows substantial voltage variation with frequency, consistent with standing waves caused by reflection at an unterminated cable end.
9. Maxima occur when incident and reflected waves reinforce; minima occur when they partially cancel. The periodic pattern can be used to estimate propagation delay and velocity factor when consecutive extrema are correctly identified.
10. A 50 Ω termination should reduce reflections compared with an open circuit. Residual variation may arise from imperfect matching, connector/adapter effects, cable attenuation, generator output settings, oscilloscope loading or reading uncertainty.

9. Conclusion and Applications

The experiment demonstrates RF transmission-line behaviour, reflections, standing waves and impedance matching. The open-circuit cable produced substantial voltage variation with frequency, indicating reflected-wave interference. A matched 50 Ω termination is expected to absorb most incident power and reduce the ripple; the remaining measured variation illustrates practical mismatch and measurement limitations.

Frequency spacing between correctly identified standing-wave extrema can be used to estimate phase velocity and velocity factor, which are important for predicting propagation delay and electrical length in coaxial cables and RF circuits. Reflection-coefficient measurements underpin return-loss and VSWR testing, while impedance matching reduces reflected power and improves power transfer.

These principles are used in antenna feed lines, radio transmitters and receivers, radar, satellite links, broadcasting, cable systems, RF test benches and wireless infrastructure. In radio astronomy, well-matched low-loss receiver connections help preserve extremely weak signals from antennas and reduce avoidable reflection ripple. Industrially, the same measurements help verify cables, connectors, loads, antennas and RF modules before system integration.